Model Kill-Switch Protocol | Design Document

From architecture to code: which classes exist, how a command moves between them, which state it can be in, who runs in parallel and what is stored.

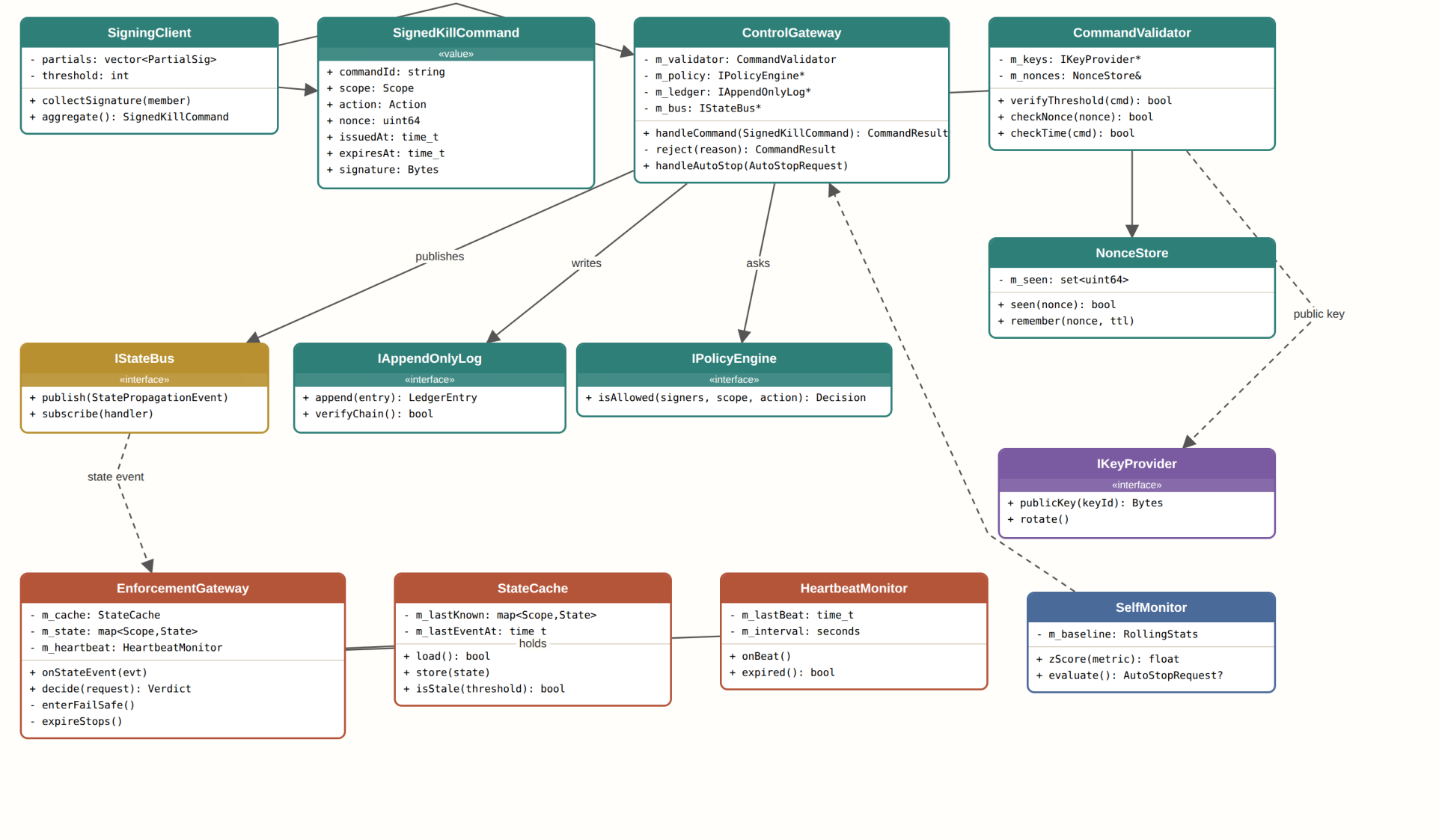
The design document comes after the architecture. The architecture answered "which components exist and who talks to whom". Here we go one level down: each component becomes classes with fields and operations, and what passes between them is defined, so code can be written without guessing.

1. Design principles

- Depend on interfaces, not implementations: everything that may be swapped (policy, log, bus, keys) is an interface, so each part can be tested alone and technology replaced without touching the gateway.
- One direction of control: commands enter only through ControlGateway. There is no bypass, so there is one place to check and defend.
- One state, several copies: the true state is set at the center and every gateway keeps a local copy. When they disagree, the gateway follows a predefined fail-safe rule.
- Every rejection is kept: a rejection is information, not noise. It is written to the security log.

2. Class diagram

Each box is a class: name on top, then fields (what it holds), then operations (what it can do). A box marked «interface» is a contract only: it fixes which operations must exist and whoever implements it decides how. A solid arrow means "uses", a dashed arrow means a message or request, and a diamond means "holds".



Class diagram

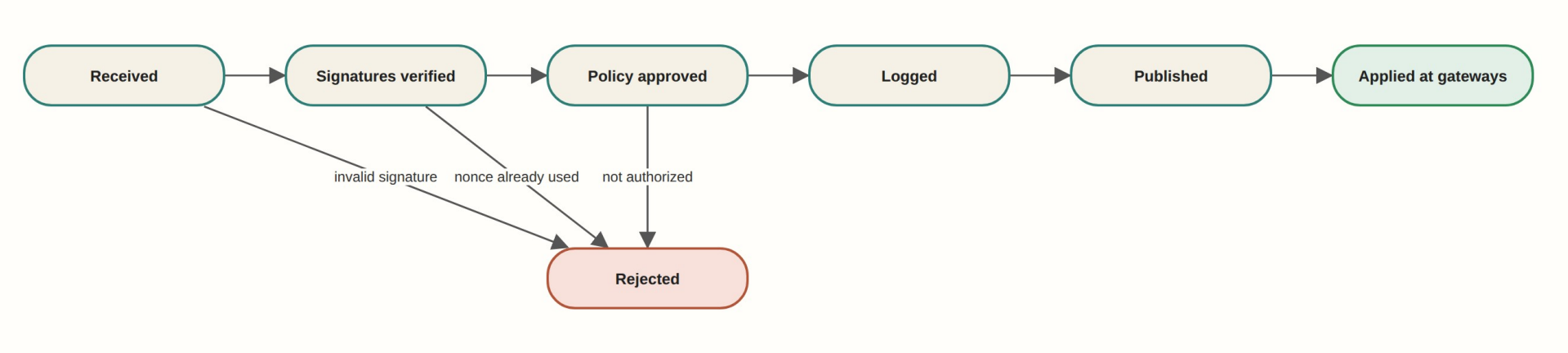
What each class does

Class	Role
SigningClient	Collects partial signatures from committee members and merges them into one command.
SignedKillCommand	Value object: the command itself, tied to a scope, an action and a time. Immutable after signing.
ControlGateway	The only door. Orchestrates the checks and the steps after them; decides nothing itself.
CommandValidator	Checks exactly three things: signature threshold, one-time nonce, time.
NonceStore	Remembers nonces already used, so an old command cannot be replayed.
IPolicyEngine	Interface to the policy engine. An implementation can change rules without touching the gateway.
IAppendOnlyLog	Interface to an append-only log. Each record points to the previous one by hash.
IStateBus	Interface to the distribution bus. Publish once and every gateway receives.
IKeyProvider	Interface to the key vault. No component holds a key itself.
EnforcementGateway	The guard next to the server. Receives state events, decides on every request, and enters fail-safe mode when the link is damaged.
StateCache	Local copy of the last known state, kept even when the network is down.
HeartbeatMonitor	Checks that heartbeats from the center keep arriving.
SelfMonitor	Computes deviation from normal and can only request a stop, through the same gateway.

3. State machines

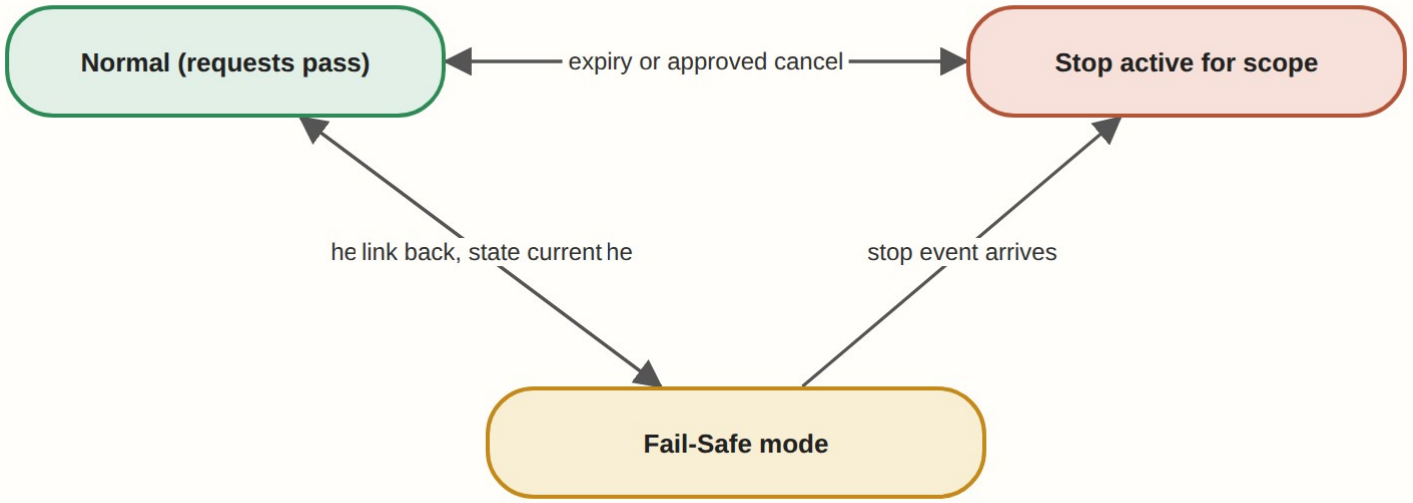
A state answers "where does this stand right now". Defining states and transitions in advance prevents undefined situations, such as a command that was published without being logged.

Lifecycle of the command



Transitions. Green is a normal end, red is a rejection, yellow is an intermediate state.

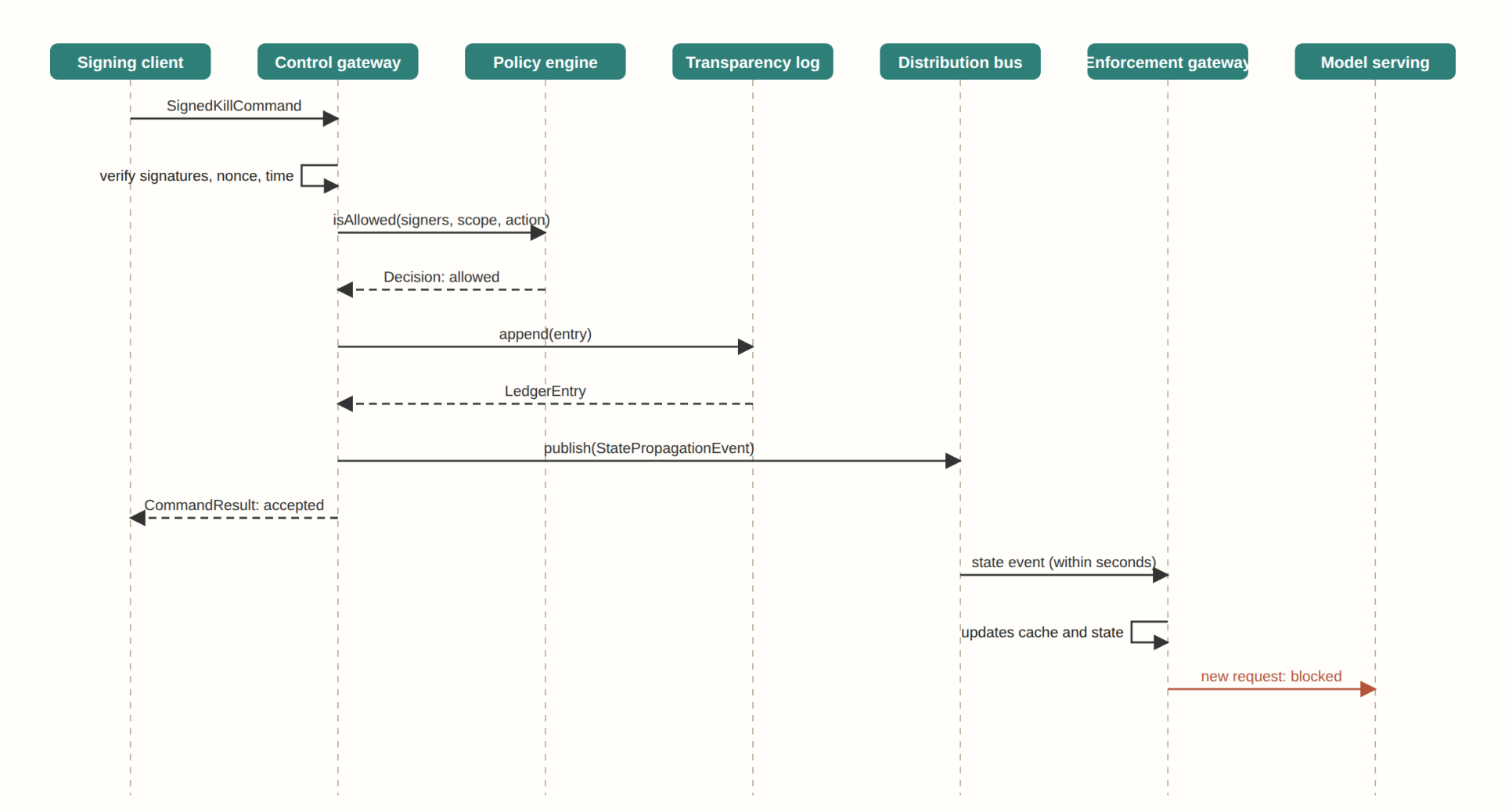
States of the enforcement gateway



States and transitions

4. Sequence diagram

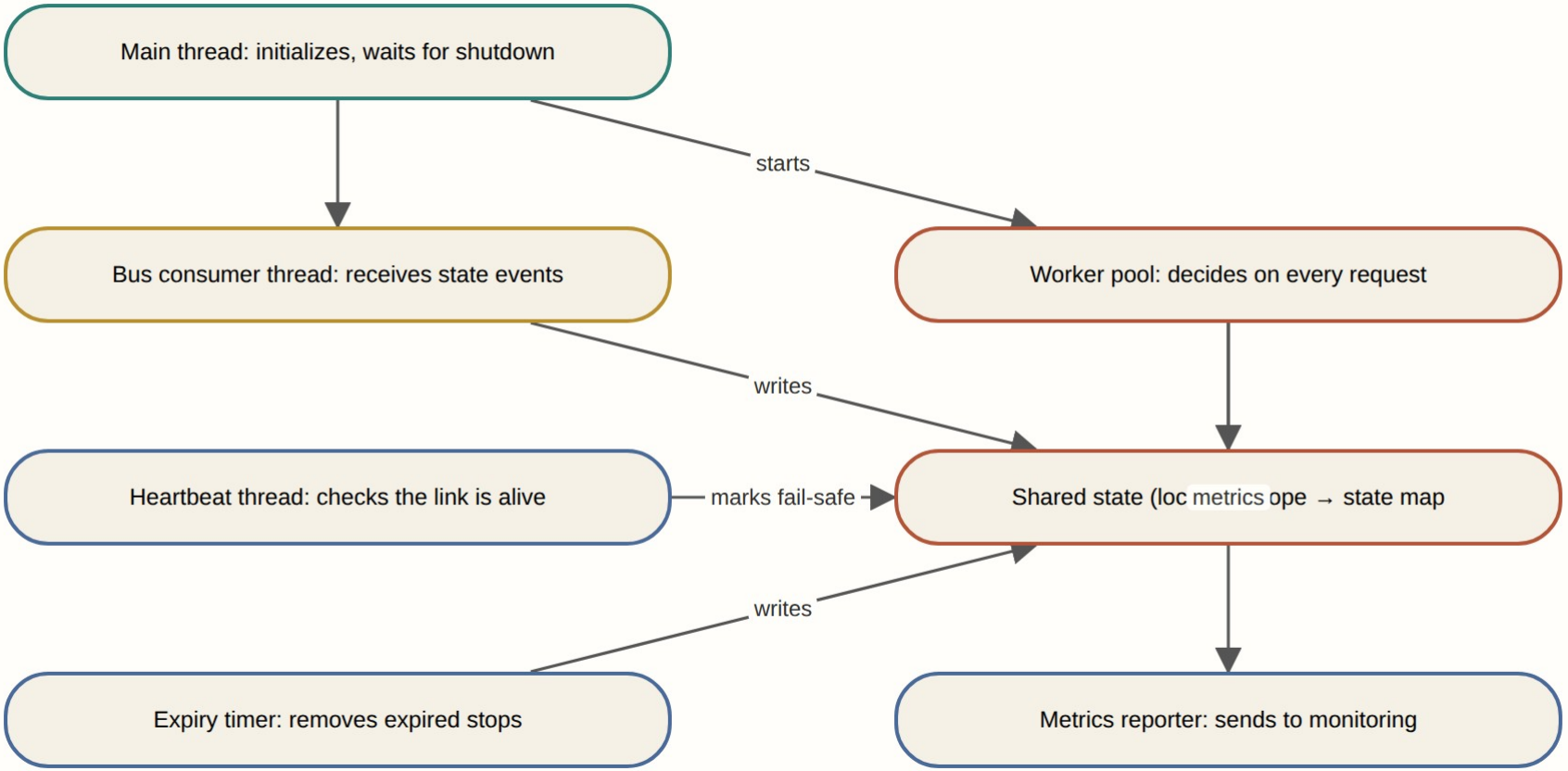
Who calls whom and in what order, top to bottom. A plain arrow is a call, a dashed arrow is a reply, and a red arrow is a blocking result.



The normal flow end to end

5. Threading model

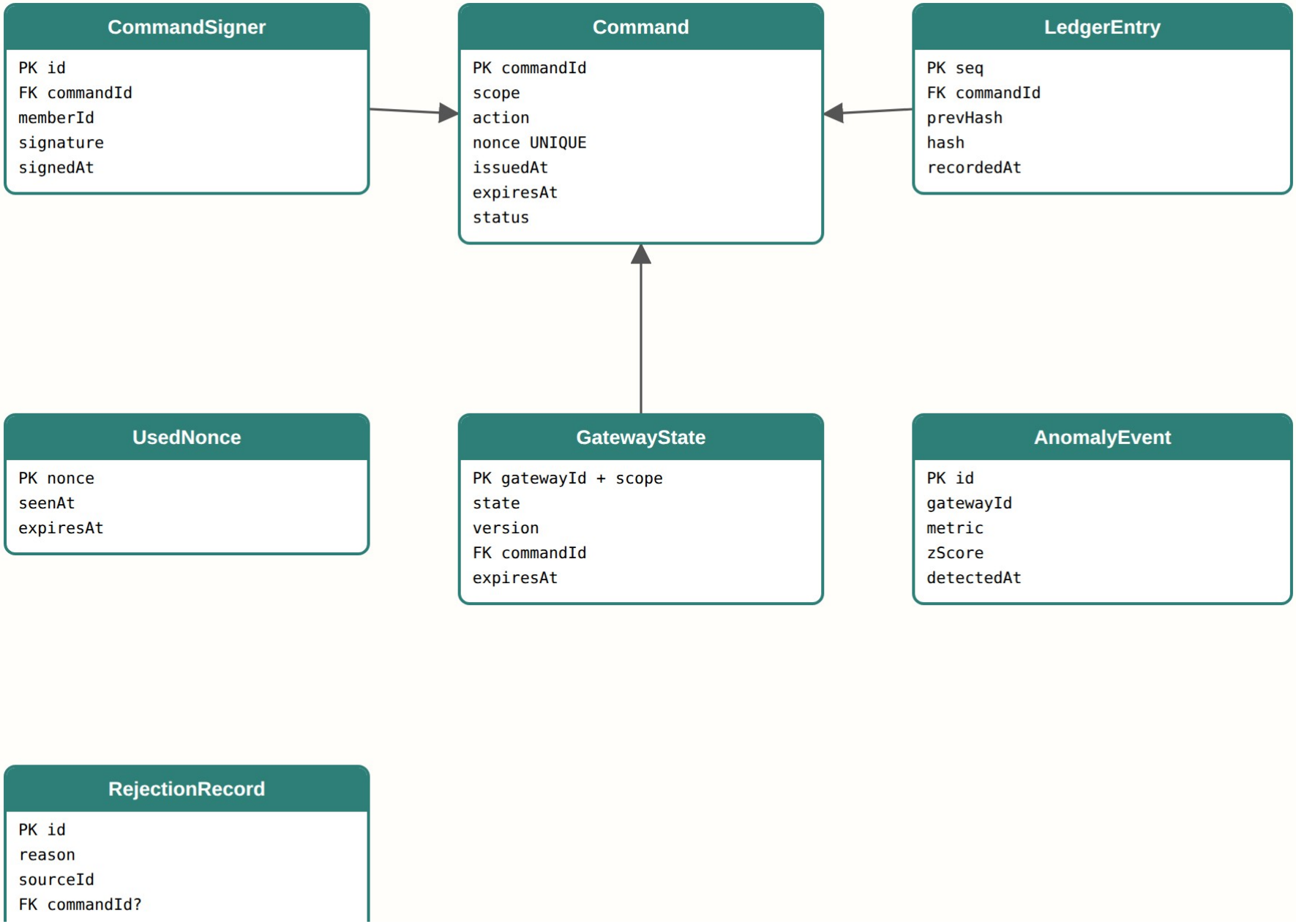
A thread is a "worker" running in parallel with others. When several threads touch the same state a lock is required, otherwise two may change it at once and leave it broken.



Threads and what they share

6. Table design

What is stored long-term and how the tables are linked. PK is a primary key that identifies a row; FK is a reference to a row in another table.


CommandSigner
PK id
FK commandId
memberId
signature
signedAt
Command
PK commandId
scope
action
nonce UNIQUE
issuedAt
expiresAt
status
LedgerEntry
PK seq
FK commandId
prevHash
hash
recordedAt
UsedNonce
PK nonce
seenAt
expiresAt
GatewayState
PK gatewayId + scope
state
version
FK commandId
expiresAt
AnomalyEvent
PK id
gatewayId
metric
zScore
detectedAt
RejectionRecord
PK id
reason
sourceId
FK commandId?

7. Implementation order by version

- **V1.0.0 Skeleton:** SigningClient, SignedKillCommand, ControlGateway and a simple enforcement gateway. One signed command stops one model.
- **V2.0.0 Verification:** CommandValidator, NonceStore, IPolicyEngine and a 3-of-5 signature threshold.
- **V3.0.0 Transparency and distribution:** IAppendOnlyLog, IStateBus and multiple regions.
- **V4.0.0 Resilience:** StateCache, HeartbeatMonitor, fail-safe mode and automatic expiry.
- **V5.0.0 Observation:** SelfMonitor, AuditSink and a central security log.